

Theory Lectures on Manufacturing Processes (MEI102)

LBM – Laser Beam Machining

(Lecture 8B)



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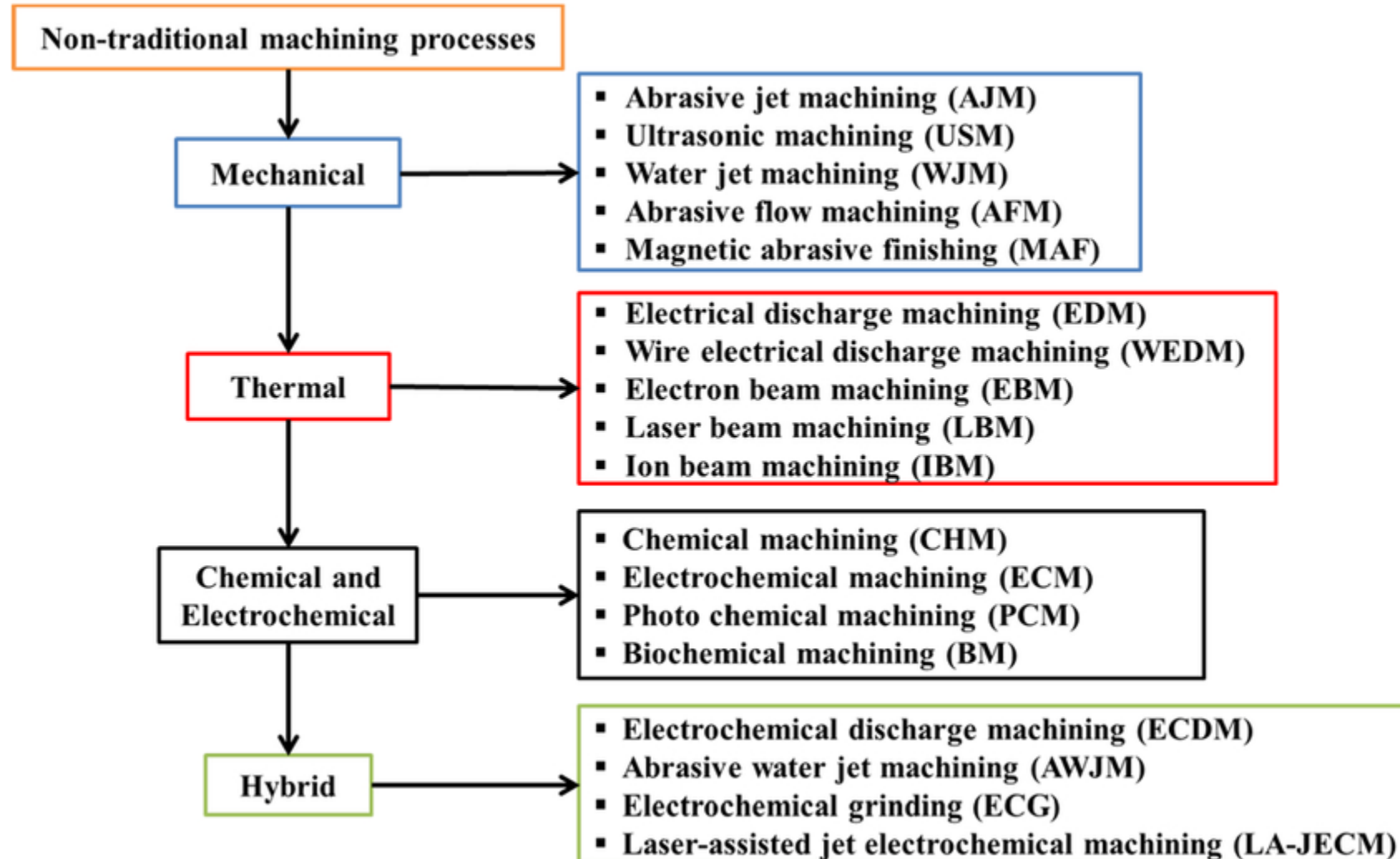
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Content prepared for theory lecture for
1st Year BTech level students for common
course on Manufacturing Processes.
Lecture duration $\approx$ 20 min



Classifications of NTM



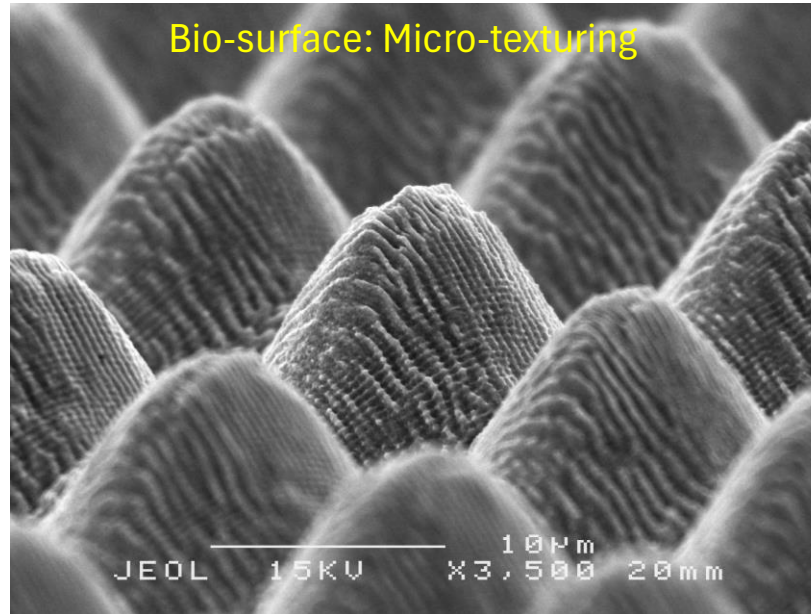


Diverse applications of laser beam

Manufacturing: Laser welding



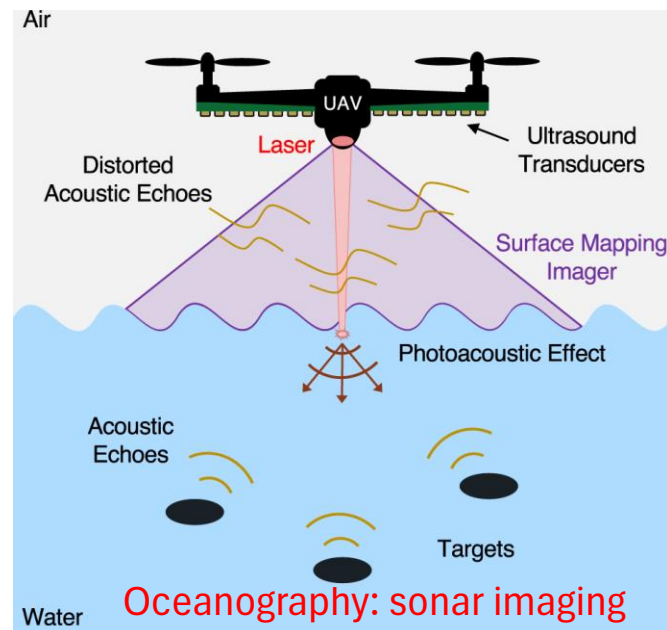
Bio-surface: Micro-texturing



Lighting & amusement: Laser lights



Medical: Laser eye treatment

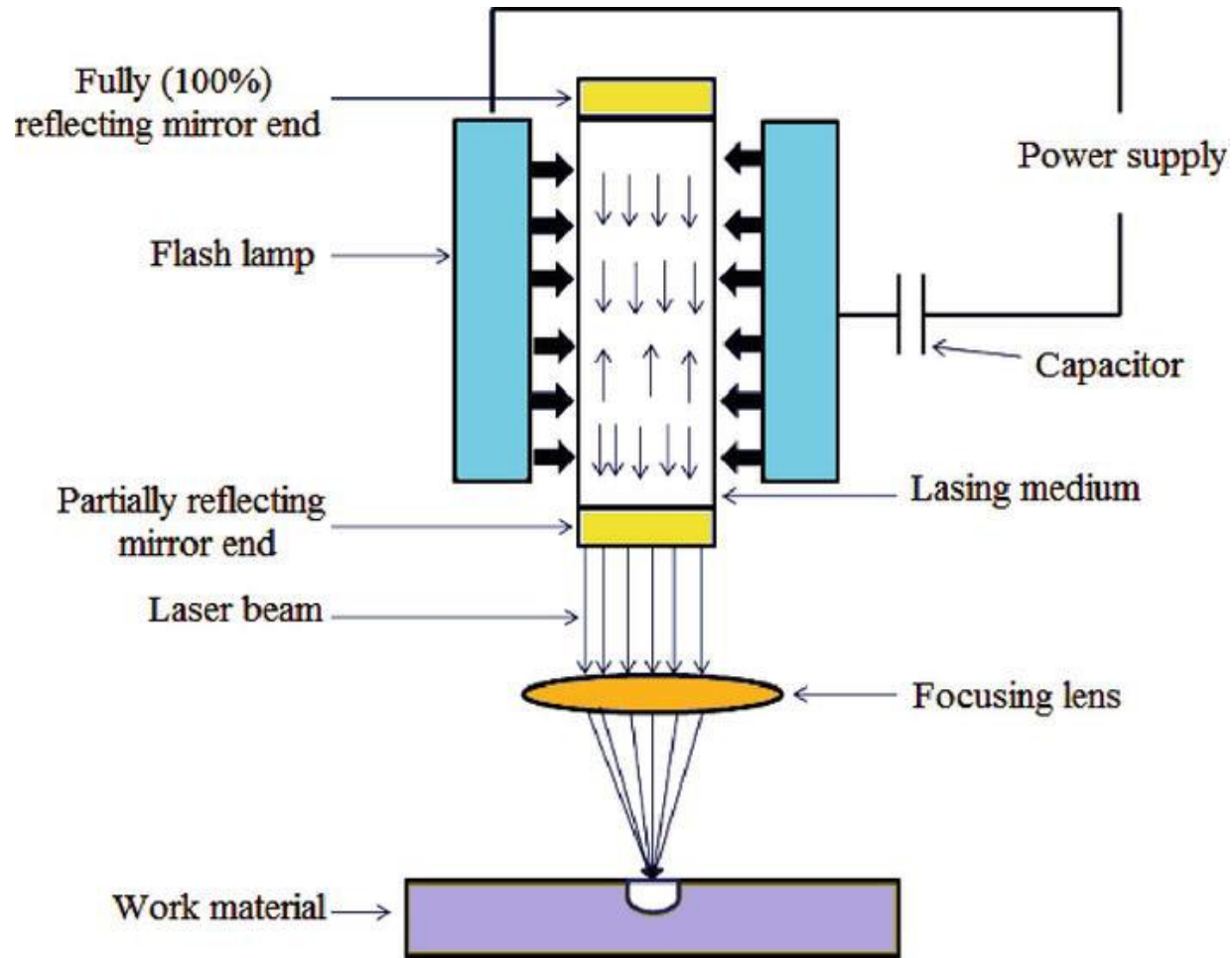


Defense: Laser weapon for long range destruction





LBM set up



Light amplification by stimulated emission of radiation

Power Supply: The power supply provides the necessary energy to generate the laser beam. It must be compatible with the specific type of laser being used and capable of delivering the required voltage range for efficient operation.

Flash Lamps: Flash lamps are gas discharge lamps which are used for generating intense light pulses. They produce intense white light that is converted into a coherent high-intensity beam.

Capacitor: Capacitors in a laser system act as energy reservoirs. They store electrical energy and release it rapidly to create the intense light pulses required for laser generation. The capacity and discharge rate of these capacitors are key factors in determining the power and frequency of the laser pulses.

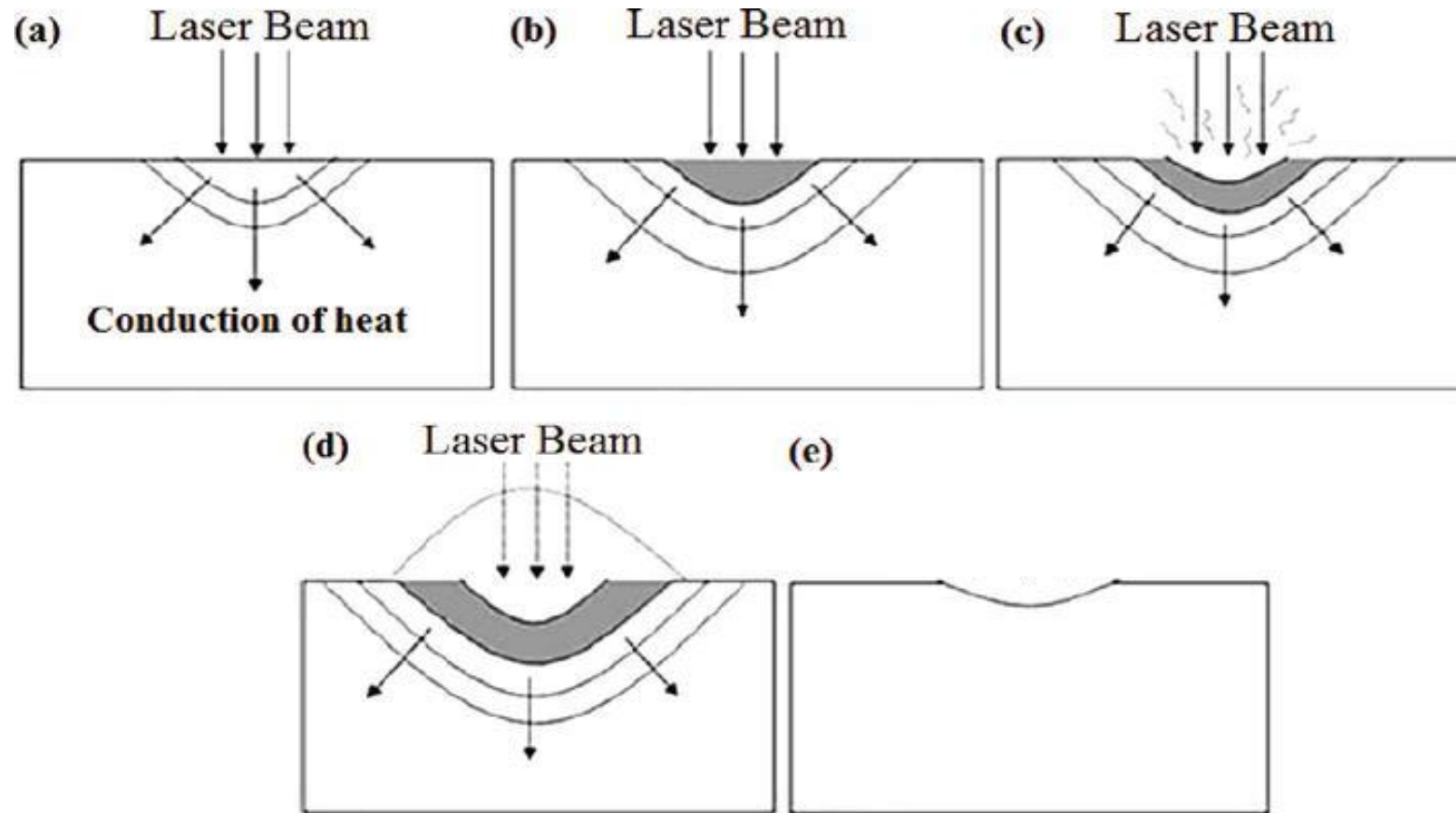
Reflecting Mirror: Reflecting mirrors are integral to directing and focusing the laser beam onto the workpiece. They must be precisely aligned and made of materials that can withstand high energy densities without degrading.

Lenses: The lenses in a laser machining system are responsible for focusing the laser beam onto the workpiece. The material and curvature of these lenses determine how the laser light is concentrated, affecting the machining quality.

Workpiece: The workpiece is the material being machined. Its properties, such as material type, reflectivity, and thermal conductivity, play a significant role in how it interacts with the laser beam. Understanding these properties is essential for optimizing the machining process and achieving desired results.



Laser ablation: Mechanism of material removal in LBM



Interactions between laser beam and work material: (a) heating, (b) localized melting of surface material, (c) vaporization of surface material, (d) formation of plasma due to interaction of intense laser beam and vaporized metal, and (e) final laser ablation mark.

Type of energy utilized: Light-assisted thermal energy

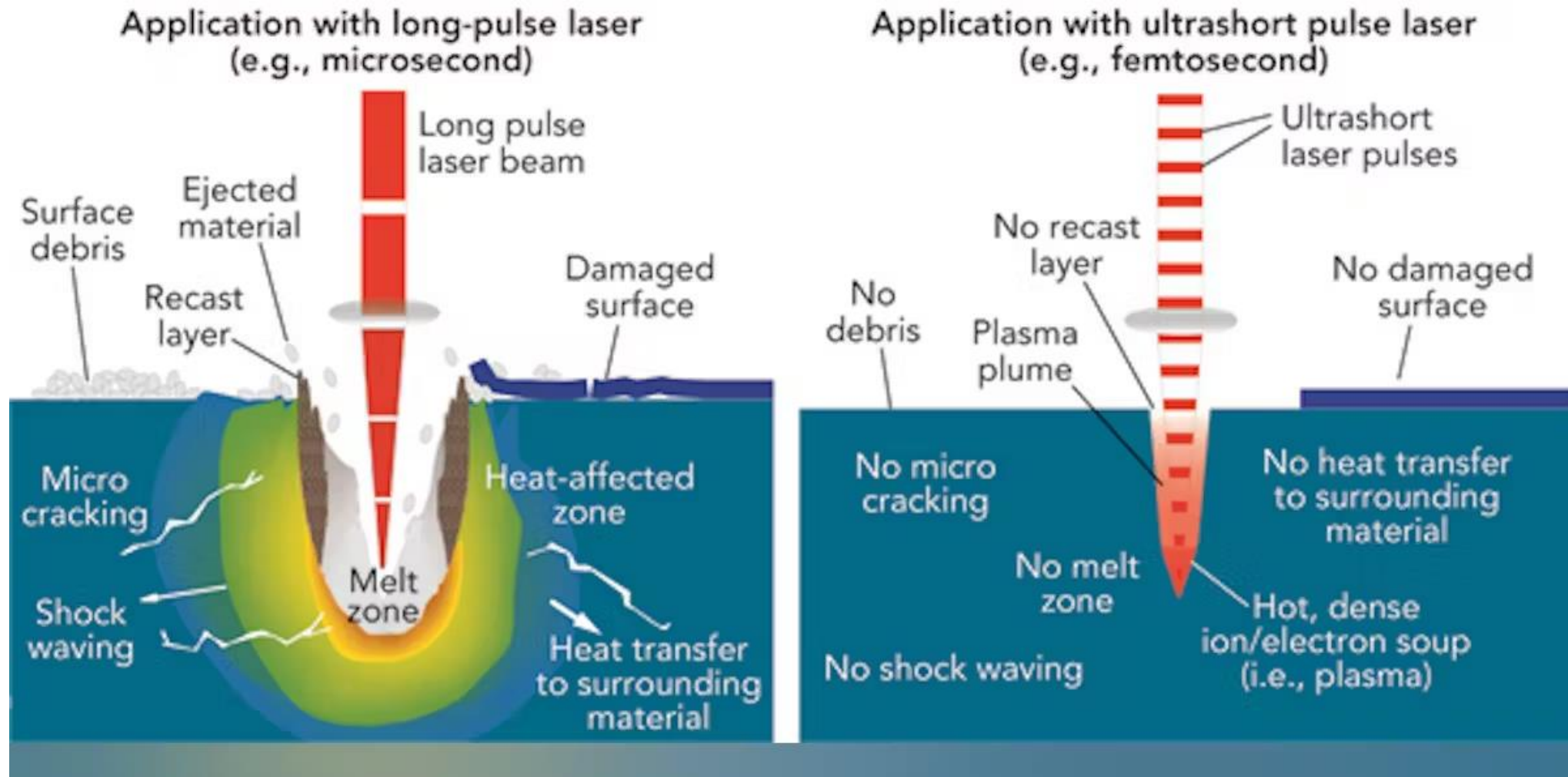
Mechanism of material removal: Laser ablation

- Laser ablation is a procedure where solid materials can be removed from the surface of that solid material upon irradiation of a laser beam. Higher flux converts the materials into plasma, whereas in presence of lower flux the material evaporates or sublimates by absorbing heat energy.

Mode of material removal: Thermal evaporation

- Note that in PAM, material removal takes place in molten and vaporized form due to plasma arc blow along with intense heat.
- In LBM, no such blow is possible. Thus liquid metal cannot be removed by laser beam. Material removal takes place only in vaporized form. The molten metals re-solidify and reside on the workpiece surface. Therefore, the process parameters of LBM are tuned in such a way that it can instantly vaporize the material with minimum melting.

Longer and shorter pulses of laser for machining



Longer pulse duration increases the heat affected zone (HAZ) and leaves high thermal stresses resulting in crack and void formation, and surface debris. Short pulse duration leads to lesser thermal conduction thus resulting in precise machining operation and good surface finish.



Advantages, disadvantages, and applications

Advantages of LBM

- It can produce excellent surface finish & complex geometry.
- LBM is a clean manufacturing technique due to less environment pollution and no requirement of chemicals.
- LBM can be easily automated for higher productivity and to achieve high speed machining.
- It uses no cutting tool; therefore no cutting forces and vibration involved (no mechanical damage but thermal damage is there).
- Wide range of work materials regardless of hardness and brittleness (from plastics to diamond) can be machined.
- Machining micro features with large aspect ratio is possible.

Disadvantages of LBM

- Machine is highly expensive - Investment cost is high.
- Optical transparent and greatly reflective materials cannot be machined using LBM
- High heat can induce thermal damage on workpiece.
- High chance of re-deposition of molten metal on machined surface.

Applications of LBM

- Cutting hard and difficult-to-machine materials.
- Drilling holes on wide range of materials.
- Micro-feature fabrication (micro-texturing) on surfaces.
- Marking and engraving on metallic surfaces.
- Laser beam can also be applied for other manufacturing applications such as welding, cladding, marking, cleaning, engraving, surface treatment, etc.



- <https://www.youtube.com/shorts/K852uV4dnSU?feature=share>
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- <https://youtu.be/mgaukC25Hqk>



Difference between LBM and PAM

Laser Beam Machining (LBM)	Plasma Arc Machining (PAM)
A high intensity beam of laser (coherent photon particles) is used to supply heat for melting and vaporizing the workpiece material.	A high temperature plasma jet (highly excited ions and electrons) is used as the heat source for melting and vaporizing the workpiece material.
No separate gas is required to supply to generate laser beam as it is directly generated using the lasing medium (Nd-YAG, Nd-Glass, Ruby, CO ₂) with the assistance of flash tube.	Suitable plasma forming gas (air, argon, nitrogen, hydrogen) is required to supply continuously to a closed gas chamber under high pressure, which ultimately comes out as high temperature plasma jet.
No electric arc is required to establish in LBM process.	An electric arc is essentially required to supply immense heat for converting the gas into plasma.
Although melting and vaporization both occur simultaneously, most of the material is removed in the form of vapor. Molten metal cannot be removed (as there is no blow possible). This molten metal re-solidifies to reside on work surface.	Both melting and vaporization occur in PAM too; however, majority of the material is blown away by the high velocity plasma jet when the workpiece material remains in molten state. A comparatively smaller fraction of the material is removed in vaporized form.
Laser beam can be allowed to strike the workpiece either continuously or intermittently (as in the case of femtosecond laser).	Plasma jet is usually allowed to continuously strike the workpiece.
In LBM, the kerf width can be maintained very small (even below 0.1 mm).	In PAM, kerf is relatively wider (3 – 10 mm).
A narrow localized heat-affected zone can be noticed in the specimens after cutting by LBM. So it causes very little thermal damage to the workpiece.	Specimens cut by PAM can have significantly wider heat-affected zone. The extent of thermal damage made in PAM is more.
LBM process capability is affected by optical properties of the workpiece surface (transparent, translucent and opaque). Highly reflective surface can reduce rate of heat input.	PAM process is not affected by the optical properties of the workpiece surface.
LBM process is not noisy and can be carried out without specific personal protections.	PAM process is very noisy. Thus proper hearing protection (like ear muffler) is needed for operators.



Comparison among various NTM processes

	Energy utilization	Mechanism of material removal	Form of material removal
AWJM	Mechanical energy	Abrasive erosion	Tiny metal particles
CHM	Chemical energy of a reaction	Chemical etching	Ions + chemical biproducts
ECM	Electro-chemical energy	Electrochemical dissolution	Ions
EDM	Thermal energy of spark	Thermal melting and vaporization	Molten and vaporized metal
PAM	Thermal energy of plasma arc	Thermal melting + plasma blow	Molten metal
LBM	Thermal energy of laser beam	Laser ablation (vaporization)	Vaporized metal